

GD32 ă°ă»¶çfšă½•ă@žă“◆ēē°ă½• ## 1. ç»@çš,, æœ¬æ¬ŧæjŧēē°ă½•ă, Ğæ¬jă@Ğæ•
'çš,, GD32 ă°ă»¶çfšă½•ăµ◆ç¬i¼ĞăĞ...ă◆«i¼š - æž,,ă°ă»¶ - æžçæµ< ST-Link -
çfšă½•ă°ă»¶ă° Flash - ä»ž Flash ē¬»ăžă°ă»¶ - æjēăĞē¬»ăžăŧ...
ă@'ă°Ğæœ¬ăœ°ă°ă»¶ă¬ă◆jă,ēēŧ æœ¬æ¬jăæ“◆ă½œă¬ē±j¼š ``text éj¹ç»@ç»ă½•i¼š...
ēš¬ç°°ŧç»@æ ŧi¼šGD32F303RCT6 / GD32F30x HD çfšă½•ă•ă...i¼šST-Link ă°ă»¶ă¬
ŧă»¶i¼šfirmware/build/gd32f303rct6_firmware.bin çfšă½•ăœ°ă◆ēi¼š0x08000000 `` ## 2.
ă°°◆ç½@æ◆jă»¶ çj@ēē@ă²ç»◆ă@°°ēŧ...ă»ă,ă•ă...i¼š ``sh brew install cmake ninja
arm-none-eabi-gcc stlink open-ocd `` çj@ēē@ă²ă»◆ă◆¬ç¬i¼š ``sh arm-none-eabi-gcc --
version cmake --version ninja --version st-info --version st-flash --version ``
çj¬ă»¶ējžăžŧē|◆æ±,i¼š | ST-Link | GD32 ă¼ăĞă◆ă¬◆| | --- | --- | SWDIO | SWDIO | |
SWCLK | SWCLK | | GND | GND | | 3.3V | 3.3V ă◆,ēēfç¬µăž< | | NRST | ă◆ēē° |
æ¬æ,,◆i¼šăž...ēj»ă...±ăœ°i¼ăĞă◆ă¬™ ST-Link ă◆ēf½ēf½ēē« Mac ē¬ŧă¬i¼ăĞă½ŧæ
— æ³•ē¬ŧă¬«ç»@æ ŧēš¬ç°°ŧăē, ## 3. ējă...ăēj¹ç»@ç»ă½•ă°æ°°šējĞēi¼š ``sh cd ... ``ă◆žç»-
ă½ă»◆ēf½ăœē¬ŧç»@ă½•ă,ă°°šējĞăē, ## 4. ç¬ă,Ğæŧi¼šăž,,ă°ă»¶ă»¶æ°°šējĞēi¼š ``sh
cmake --build firmware/build `` æœ¬æ¬jă@žē™...ē³¼ăŧŧi¼š ``text ninja: no work to do. `` ē
æ¬ži¼š - æž,,ă°ç³»»ŧæŧæŧŧă◆ăžă◆çžă½ă°ă°ă»¶ă²ç»◆ă¬æœæ¬°çš,,ăē, -
ă◆¬ç¬æžă½jç¬çžæœ°ă°šç°°@ējējĞçfšă½•ăē, ă½ă°ă°◆çfšă½•ă½jç¬çš,,æ¬ŧă»¶i¼š
``text firmware/build/gd32f303rct6\_firmware.bin `` ăj,æžœç¬ă,Ğæ¬jăž,,ă°ă¬`firmware/
build` ă,ăăă¬i¼ăĞēēē|ă...ă°°šējĞă@Ğæ¬é...◆ç½@ă½ă»◆i¼š ``sh cmake -S
firmware -B firmware/build -G Ninja -DCMAKE_TOOLCHAIN_FILE=.../firmware/cmake/arm-
none-eabi-gcc.cmake -DCMAKE_BUILD_TYPE=Release cmake --build firmware/build `` ##
5. ç¬ă¬ăĞæŧi¼šăžçæµ< ST-Link ă°Ğç»@æ ŧēš¬ç°°ŧæ°°šējĞēi¼š ``sh st-info --probe ``
æœ¬æ¬jă@žē™...ē³¼ăŧŧi¼š ``text Found 1 stlink programmers version: V2J37S7 serial:
XXXXXXXXXXXX flash: 262144 (pagesize: 2048) sram: 65536 chipid: 0x414 dev-type:
F1xx_HD `` ç»æžœēšŧēŧi¼š - `Found 1 stlink programmers`i¼šMac ă²ē¬ŧă¬ă° 1 ă,ă ST-
Linkăē, - `flash: 262144`i¼šç»@æ ŧēš¬ç°°ŧ Flash ă° 256 KiBăē, - `sram: 65536`i¼šēj™æ¬
ST-Link ēēšējŧ STM32F1 ă...¼ă@'ă◆◆ēē@ăšŧăšçš,, SRAM
ăjăæ¬i¼ăĞă,ă½œă,ēē¼ăžŧē,,šæœ¬ă¼ă◆ăē@ăē, - `dev-type: F1xx\_HD`i¼šstlink ă½jç¬
STM32F1 ēēă¬ŧăŧă...¼ă@'ă◆◆ēē@ē¬ŧă¬ēŧŧ GD32 ēš¬ç°°ŧăē, æ¬æ,,◆i¼šăžç»-
ē°fē•çj@ēē@æœ¬ăējăžŧăž, Keil ăŧç¬ăŧæ¬ž `IRAM(0x20000000,0x0C000)`i¼ăĞă◆ 48
KiB ă◆¬ç¬ SRAMăē,ēē¼ăžŧē,,šæœ¬ă°ă½jç¬` RAM LENGTH =
48K`i¼ăĞă◆ă¬™æ¬ējă¼šēē½ă¬œ— æ¬ăœ°ă◆ēăjăœ`SystemInit() ă°ă◆ăžēšjăă
HardFaultăē, çœă°ēj™ă,ăē¼ăŧŧă¬ži¼ăĞă¬ăŧç»šç»çfšă½•ăē
, ăj,æžœæ¬jăœ°°çœă°ç»@æ ŧēš¬ç°°ŧăjăæ¬i¼ăĞă...ăŧæŧæŧŧi¼š -
ă¼ăĞă◆ă¬ă¬ă¬ă²ç¬µăē, - SWDIO/SWCLK æ¬ă◆ăžŧă¬ăē, - ST-Link
ă°Ğă¼ăĞă◆ă¬ă¬ă¬ă...±ăœ°ăē, - USB ç°jă¬ă¬ă¬ă¬ă¬ă°ă¬ă¼ăē¼ăē, - ăj...ējăœ
—ējăžăžŧŧ NRST æ¬æœ°ă½◆ă¬ă½ăŧăžçæµăē, ## 6. ç¬ă,°æŧi¼šçfšă½•ă°ă»¶ă»¶
æ°°šējĞēi¼š ``sh st-flash --reset write firmware/build/gd32f303rct6_firmware.bin 0x08000000
`` ă½ă»◆ă¬ă°i¼š - `st-flash`i¼šă½jç¬` ST-Link æ°°šējĞē Flash æ¬ă½œăē, - --
reset`i¼šăŧŧă...ă@Ğæ¬ă◆ăžă¬ă½◆ç»@æ ŧēš¬ç°°ŧăē, - `write`i¼šăŧŧă...ă¬
ŧă»¶ă¬ēš¬ç°°ŧ Flashăē, - `firmware/build/
gd32f303rct6_firmware.bin`i¼šējçfšă½•çš,,ă°Ğējă¬ăŧă»¶ăē, - `0x08000000`i¼šGD32 ăŧ...
ēf¬ Flash ēµăŧăœ°ă◆ăē, æœ¬æ¬jă@žē™...ē³¼ăŧŧi¼š ``text st-flash 1.8.0
2026-05-15T00:07:43 INFO common.c: F1xx_HD: 64 KiB SRAM, 256 KiB flash in at least 2
KiB pages. file firmware/build/gd32f303rct6_firmware.bin md5 checksum:
a7ef2964b7252eefa5a451875d88bd, stlink checksum: 0x0003f6fc 2026-05-15T00:07:43 INFO
common_flash.c: Attempting to write 3004 (0xbbc) bytes to stm32 address: 134217728
(0x8000000) -> Flash page at 0x8000000 erased (size: 0x800) -> Flash page at 0x8000800 erased
(size: 0x800) 2026-05-15T00:07:43 INFO flash_loader.c: Starting Flash write for VL/F0/F3/
F1_XL 2026-05-15T00:07:43 INFO flash_loader.c: Successfully loaded flash loader in sram
2026-05-15T00:07:43 INFO flash_loader.c: Clear DFSR 2026-05-15T00:07:43 INFO
flash_loader.c: Clear CFSR 2026-05-15T00:07:43 INFO flash_loader.c: Clear HFSR 1/2 pages
written 2/2 pages written 2026-05-15T00:07:43 INFO common_flash.c: Starting verification of
write complete 2026-05-15T00:07:43 INFO common_flash.c: Flash written and verified! jolly

good! 2026-05-15T00:07:43 INFO common.c: NRST is not connected --> using software reset via AIRCR `` `ä...³é”@ç»“æžœĩ¼š ``text Flash written and verified! jolly good! ``
è¿™ëĵčđŧ¼š - Flash ä²æ“ĳē™ðā€,- ā°ă»¶ā²â†™ā...¥ā€, - â†™ā...¥ā◆Ž stlink
â²ê‡ªŠ̄eªCēē ◆æ◆āŚŸā€, ā...³ºŽē¿™æ◆jæ◆◆čđŧ¼š ``text NRST is not connected -->
using software reset via AIRCR `` `è¿™ā,◆æ~ çfşå½•ā±ē ¥ā€,ā®fėjčđª½`ā%◆ ST-Link
æ²jæœ%œ¿žæŻȲçĭ-ā»¶āð◆ā½◆ç◆¿ `NRST`ĩ¼Çæ%œā»¥ā-¥ā...·å½¿ç”ē½-ā»¶āð◆ā½◆ā€
,ā◆ªēĳā%œ◆é◆çā‡çŽ° `Flash written and verified`ĩ¼Ççfşå½•ā±æ~ æ◆āŚŸçš,,ā€, ## 7.
ç—ā»æ¥ĩ¼šē~»āż Flash ā,ª°†è¿)ā,€æ¥çĳ@è®ðēŠ~ç%o‡ā,çš,,â†...
â®¹ā`Çæœ-āœ°ā°ă»¶ā®Çā...¨ā,€è‡ ĩ¼Çæ%œ§ejCē~»āžĩ¼š ``sh st-flash read firmware/
build/gd32f303rct6_firmware_readback.bin 0x08000000 3004 `` ā½ă»ðā◆«ä¹%œĩ¼š -
read`ĩ¼šă»ŽēŠ~ç%o‡ Flash ē~»ā◆-â†...â®¹ā€, - `firmware/build/
gd32f303rct6_firmware_readback.bin`ĩ¼šē~»āžă;◆ā çš,,æ-‡ă»¶ā€, - `0x08000000`ĩ¼šē~»ā◆-
ēm.āš«āœ°ā◆ēā€, - `3004`ĩ¼šē~»ā◆-é•¿ā°ĩ¼Çā◆•ā½◆ā—
ēŠ,ĩ¼Çā¹ā”ª½`ā%œā°ă»¶ā®ðşå½◆ā€, æœ-æ¬jā®žē™...ē¾“ā‡°ĩ¼š ``text st-flash 1.8.0
2026-05-15T00:07:57 INFO common.c: F1xx_HD: 64 KiB SRAM, 256 KiB flash in at least 2
KiB pages. 2026-05-15T00:07:57 INFO common.c: read from address 0x08000000 size 3004 ``
ē~»āžæ-‡ă»¶ĩ¼š ``text firmware/build/gd32f303rct6\_firmware\_readback.bin `` ## 8. ç—ā”æ-
¥ĩ¼šæ~”ē¾ƒæœ-āœ°ā°ă»¶ā`Çē~»āžă°ă»¶ æ%œ§ejČĩ¼š ``sh cmp firmware/build/
gd32f303rct6_firmware.bin firmware/build/gd32f303rct6_firmware_readback.bin ``
æœ-æ¬jā®žē™...ç»“æžœĩ¼š ``text æ— è¾“ā‡° `` ē~æ~Žĩ¼š - `cmp`
æ²jæœ%œē¾“ā‡°ĩ¼Çēĳčđª,ðā,ªæ-‡ă»¶ā®Çā...¨ā,€è‡ ā€, - āĴ,æžœæ-‡ă»¶ā,◆ā◆Čēĩ¼Ç`cmp`
ā¼šē¾“ā‡°ç—ā,€ā,¨ā,◆ā◆Çā—ēŠ,çš,,ā½◆ç½½®ā€, ## 9. ç—ā...æ¥ĩ¼šē®jç®— SHA256
æ%œ§ejČĩ¼š ``sh shasum -a 256 firmware/build/gd32f303rct6_firmware.bin firmware/build/
gd32f303rct6_firmware_readback.bin `` æœ-æ¬jā®žē™...ē¾“ā‡°ĩ¼š ``text
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware.bin
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware_readback.bin `` ā,ðā,ª SHA256 ā®Çā...¨ā,€è‡ ĩ¼Çē~æ~Žĩ¼š -
æœ-āœ°ā°ă»¶æ-‡ă»¶ā`ÇēŠ~ç%o‡ Flash ē~»āžæ-‡ă»¶ā®Çā...¨ā,€è‡ ā€, -
æœ-æ¬jçfşå½•ç»“æžœā◆āžæ° ā€◆ā◆ ēªCēē ◆ā€, ## 10. æœçYçfşå½•ā½ă»ð
āĴ,æžœā◆ªæ~æ-¥ā,,ā¼ā◆ā◆ā◆Žā¿«ēČYçfşå½•ĩ¼Çā½¿ç”¨ā,€é◆çā,ðæ◆jā½ă»ðā◆³ā◆ ĩ¼š
``sh cd ... st-flash --reset write firmware/build/gd32f303rct6\_firmware.bin 0x08000000 ``
āĴ,æžœă»Łç ◆æœ%œ”āŚ~ĩ¼Ççfşå½•ā%œ◆ā...æž,,ā”ĩ¼š ``sh cmake --build firmware/build
st-flash --reset write firmware/build/gd32f303rct6_firmware.bin 0x08000000 `` ## 11.
æœ-æ¬jç»“ē® æœ-æ¬jā®Çæ°æµ◆ç~ç»“ē®ĩ¼š - ā°ă»¶æž,,ā”çš¶æ€◆æŁā,,ā€, - ST-Link
æŁā,,ē~†ā`ā«ā€, - ç®æ †ēŠ~ç%o‡æŁā,,ē~†ā`ā«ā€, - ā°ă»¶æ~◆āŚŸâ†™ā...¥ `0x08000000`ā€, -
`st-flash` ē‡ªās`æ jēªCæ~◆āŚŸā€, - æ%œ«āŚ~ē~»āžæ-‡ă»¶ā`ÇāŽŸāš«.`bin` ā®Çā...¨ā,€è‡ ā€, -
- SHA256 æ jēªÇā,€è‡ ā€, ā æđĩ¼Çæœ-æ¬j GD32 ā°ă»¶¶çfşå½•æ◆āŚŸā€, ## 12. ā◆Žç»-
çŽ°āœ°ēªCēē ◆ çfşå½•ā®Çæ~◆ā◆Žĩ¼Çæ~¿ā◆ā,Šā”ē¿ējČēā½`ā%œ◆čđª¾ā°ă»¶ĩ¼š -
RGB LED ā½¿ç”” PC0`ā€◆PC1`ā€◆PC2`ā€, - ā,%œā,ª LED æ~◆ 250 ms` ēĳā°◆ç¿»ē½-ā€
- , - USART0 ā½¿ç”” PA9 / PA10 ĩ¼Çæ³çç%œ'çŽ‡ 115200 8N1`ā€, -
ā,²ā◆Łā¼šē¾“ā‡ªā◆āŚ~ā¿jæ◆-ā`Çā”æœYæçš`tick`ā€
, āĴ,æžœéœēĳ◆ēš,ā~Yā,²ā◆łĩ¼Çē¿žæŽ¥ USB-TTLĩ¼š | USB-TTL | GD32 ā¼ā◆«æ◆¿ || --- |
--- || RX | PA9 / USART0_TX || TX | PA10 / USART0_RX || GND | GND |
ā,²ā◆Łæ%œ“ā¼āçđª¾āĩ¼š ``sh screen /dev/tty.usbserial-xxxx 115200 ``